Algorithm 1 PSO for QCAE Hyperparameter Optimization

Require: Training split D_{train}, validation split D_{val}, population size N, iterations T, bounds $[0,1]^4$, fitness function $\mathcal{F}$, inertia weight w, cognitive coefficient c_1, social coefficient c_2

Ensure: Best hyperparameter vector x^*

Initialize positions $\{x_i\}_{i=1}^{N}$ uniformly over $[0,1]^4$; set velocities $v_i \leftarrow 0$
Train and evaluate each x_i via $\mathcal{F}$; set personal bests $p_i \leftarrow x_i$
Set global best $x^* \leftarrow p_i$ for the i minimizing $\mathcal{F}(p_i)$
for $t = 1$ to T **do**
    for each particle $i = 1, \ldots, N$ **do**
        Draw random $r_1, r_2 \sim U(0,1)^4$
        Update velocity: $v_i \leftarrow w\, v_i + c_1 r_1 \odot (p_i - x_i) + c_2 r_2 \odot (x^* - x_i)$
        Update position: $x_i \leftarrow \text{clip}(x_i + v_i, [0,1]^4)$
    end for
    for each particle $i = 1, \ldots, N$ **do**
        Decode x_i, train the network, and evaluate $\mathcal{F}(x_i)$
        if $\mathcal{F}(x_i) < \mathcal{F}(p_i)$ **then**
            Update personal best $p_i \leftarrow x_i$
        end if
    end for
    Update global best $x^* \leftarrow p_i$ for the i minimizing $\mathcal{F}(p_i)$, if improved
end for
return x^*